DBMS Lab Assignment-1 — Solutions

Q1: Display the data in existing default employees or emp or employee table.

SELECT * FROM employees; -- OR: SELECT * FROM emp; -- OR: SELECT * FROM employee;

Output:

```
Note: Output depends on the pre-existing table in your schema (default Oracle sample table EMP is
shown here):


EMPNO ENAME      JOB       MGR HIREDATE   SAL  COMM DEPTNO
----- ---------- --------- --- --------- ---- ----- ------
 7369 SMITH      CLERK    7902 17-DEC-80  800          20
 7499 ALLEN      SALESMAN 7698 20-FEB-81 1600   300    30
 7521 WARD       SALESMAN 7698 22-FEB-81 1250   500    30
 7566 JONES      MANAGER  7839 02-APR-81 2975          20
 7654 MARTIN     SALESMAN 7698 28-SEP-81 1250  1400    30
 7698 BLAKE      MANAGER  7839 01-MAY-81 2850          30
 7782 CLARK      MANAGER  7839 09-JUN-81 2450          10
 7788 SCOTT      ANALYST  7566 09-DEC-82 3000          20
 7839 KING       PRESIDENT     17-NOV-81 5000          10
 7844 TURNER     SALESMAN 7698 08-SEP-81 1500     0    30
 7876 ADAMS      CLERK    7788 12-JAN-83 1100          20
 7900 JAMES      CLERK    7698 03-DEC-81  950          30
 7902 FORD       ANALYST  7566 03-DEC-81 3000          20
 7934 MILLER     CLERK    7782 23-JAN-82 1300          10
14 rows selected.
```

Q2: Display the structure of the above table.

DESC employees; -- OR: DESCRIBE employees;

Output:

```
Name      Null? Type
--------- ----- -------------
EMPNO     NOT NULL NUMBER(4)
ENAME              VARCHAR2(10)
JOB                VARCHAR2(9)
MGR                NUMBER(4)
HIREDATE           DATE
SAL                NUMBER(7,2)
COMM               NUMBER(7,2)
DEPTNO             NUMBER(2)
```

Q3: Create a table "Student" with the following structure: Roll Number(5), Name Varchar2(30), Age Number(5), Course Varchar2(5), Math Number(6,2), Physics Number(6,2), Computer Number(6,2), Birthday Date.

CREATE TABLE Student (Roll NUMBER(5), Name VARCHAR2(30), Age NUMBER(5), Course VARCHAR2(5), Math NUMBER(6, 2), Physics NUMBER(6, 2), Computer NUMBER(6, 2), Birthday DATE);

Output:

```
Table created.
```

```
DESC Student;
```

```
Name     Null? Type
-------- ----- -----------
ROLL           NUMBER(5)
NAME           VARCHAR2(30)
AGE            NUMBER(5)
COURSE         VARCHAR2(5)
MATH           NUMBER(6,2)
PHYSICS        NUMBER(6,2)
COMPUTER       NUMBER(6,2)
BIRTHDAY       DATE
```

Q4: Create table MSc from the Student table with the same fields and same structure but without any data.

CREATE TABLE MSc AS SELECT * FROM Student WHERE 1 = 0;

Output:

```
Table created.
```

Q5: Display the structure of MSc table.

DESC MSc; -- OR: DESCRIBE MSc;

Output:

```
DESC MSc;
```

```
Name     Null? Type
-------- ----- -----------
ROLL           NUMBER(5)
NAME           VARCHAR2(30)
AGE            NUMBER(5)
COURSE         VARCHAR2(5)
MATH           NUMBER(6,2)
PHYSICS        NUMBER(6,2)
COMPUTER       NUMBER(6,2)
BIRTHDAY       DATE
```

Q6: Create table MCA from the Student table with the same fields and same structure but without any data. Rename Course → Department and Name → First_Name.

CREATE TABLE MCA AS SELECT Roll, Name AS First_Name, Age, Course AS Department, Math, Physics, Computer, Birthday FROM Student WHERE 1 = 0;

Output:

```
Table created.
```

Q7: Display the structure of MCA table.

DESC MCA; -- OR: DESCRIBE MCA;

Output:

```
DESC MCA;
Name       Null? Type
---------- ----- -----------
ROLL              NUMBER(5)
FIRST_NAME        VARCHAR2(30)
AGE               NUMBER(5)
DEPARTMENT        VARCHAR2(5)
MATH              NUMBER(6,2)
PHYSICS           NUMBER(6,2)
COMPUTER          NUMBER(6,2)
BIRTHDAY          DATE
```

Q8: Insert the following records into the Student table:

INSERT INTO Student VALUES (1, 'Rahul', 19, 'BCA', 79.5, 67, 89, '15-jun-93'); INSERT INTO Student VALUES (2, 'Kunal', 21, 'BCA', 68, 76, 59.5, '16-aug-91'); INSERT INTO Student VALUES (3, 'Aditi', 20, 'MSc', 90, 73, 56, '20-sep-92'); INSERT INTO Student VALUES (4, 'Sumit', 20, 'MCA', 57.5, 78, 81, '07-dec-91'); INSERT INTO Student VALUES (5, 'Anirban', 22, 'MCA', 80, 68, 63, '15-sep-94'); INSERT INTO Student VALUES (6, 'Kumkum', 21, 'BCA', 72, 54.5, 60, '08-feb-95'); INSERT INTO Student VALUES (7, 'Suman', 21, 'BCA', 91.5, 32, 61, '10-mar-94'); INSERT INTO Student VALUES (8, 'Rohit', 22, 'MSc', 85, 76, 92, '19-apr-92');

Output:

```
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
```

Q9: Display all the students details from Student table.

SELECT * FROM Student;

Output:

```
ROLL NAME     AGE COURSE MATH PHYSICS COMPUTER BIRTHDAY
```

```
---- -------- --- ------ ---- ------- -------- --------
   1 Rahul     19 BCA    79.5      67       89 15-JUN-93
   2 Kunal     21 BCA      68      76     59.5 16-AUG-91
   3 Aditi     20 MSc      90      73       56 20-SEP-92
   4 Sumit     20 MCA    57.5      78       81 07-DEC-91
   5 Anirban   22 MCA      80      68       63 15-SEP-94
   6 Kumkum    21 BCA      72    54.5       60 08-FEB-95
   7 Suman     21 BCA    91.5      32       61 10-MAR-94
   8 Rohit     22 MSc      85      76       92 19-APR-92
8 rows selected.
```

Q10: Find out the details of the student with Roll no 5 from Student table.

SELECT * FROM Student WHERE Roll = 5;

Output:

```
ROLL NAME     AGE COURSE MATH PHYSICS COMPUTER BIRTHDAY
---- -------- --- ------ ---- ------- -------- --------
   5 Anirban   22 MCA      80      68       63 15-SEP-94
1 row selected.
```

Q11: Show the Roll, Name, marks of all subjects for all students.

SELECT Roll, Name, Math, Physics, Computer FROM Student;

Output:

```
ROLL NAME     MATH PHYSICS COMPUTER
---- -------- ---- ------- --------
   1 Rahul    79.5      67       89
   2 Kunal      68      76     59.5
   3 Aditi      90      73       56
   4 Sumit    57.5      78       81
   5 Anirban    80      68       63
   6 Kumkum     72    54.5       60
   7 Suman    91.5      32       61
   8 Rohit      85      76       92
8 rows selected.
```

Q12: Display all the student details belonging to the BCA department.

SELECT * FROM Student WHERE Course = 'BCA';

Output:

```
ROLL NAME     AGE COURSE MATH PHYSICS COMPUTER BIRTHDAY
```

```
---- -------- --- ------ ---- ------- -------- --------
   1 Rahul     19 BCA    79.5      67       89 15-JUN-93
   2 Kunal     21 BCA      68      76     59.5 16-AUG-91
   6 Kumkum    21 BCA      72    54.5       60 08-FEB-95
   7 Suman     21 BCA    91.5      32       61 10-MAR-94
4 rows selected.
```

Q13: Insert data in the MCA table from Student table where course is MCA. (Note: MCA table has renamed columns — Name→First_Name, Course→Department)

INSERT INTO MCA (Roll, First_Name, Age, Department, Math, Physics, Computer, Birthday) SELECT Roll, Name, Age, Course, Math, Physics, Computer, Birthday FROM Student WHERE Course = 'MCA';

Output:

```
2 rows created.

SELECT * FROM MCA;

ROLL FIRST_NAME AGE DEPARTMENT MATH PHYSICS COMPUTER BIRTHDAY
---- ---------- --- ---------- ---- ------- -------- --------
   4 Sumit       20 MCA        57.5      78       81 07-DEC-91
   5 Anirban     22 MCA          80      68       63 15-SEP-94
2 rows selected.
```

Q14: Insert data in the MSc table from Student table where course is MSc.

INSERT INTO MSc SELECT * FROM Student WHERE Course = 'MSc';

Output:

```
2 rows created.

SELECT * FROM MSc;

ROLL NAME     AGE COURSE MATH PHYSICS COMPUTER BIRTHDAY
---- -------- --- ------ ---- ------- -------- --------
   3 Aditi     20 MSc      90      73       56 20-SEP-92
   8 Rohit     22 MSc      85      76       92 19-APR-92
2 rows selected.
```

Q15: Display the structure of Student table.

DESC Student; -- OR: DESCRIBE Student;

Output:

```
DESC Student;
Name     Null? Type
-------- ----- -----------
ROLL             NUMBER(5)
NAME             VARCHAR2(30)
AGE              NUMBER(5)
COURSE           VARCHAR2(5)
MATH             NUMBER(6,2)
PHYSICS          NUMBER(6,2)
COMPUTER         NUMBER(6,2)
BIRTHDAY         DATE
```

Q16: Display all the details of each student in Student table with Course appearing first in the display column list.

SELECT Course, Roll, Name, Age, Math, Physics, Computer, Birthday FROM Student;

Output:

```
COURSE ROLL NAME     AGE MATH PHYSICS COMPUTER BIRTHDAY
------ ---- -------- --- ---- ------- -------- --------
BCA       1 Rahul     19 79.5      67       89 15-JUN-93
BCA       2 Kunal     21   68      76     59.5 16-AUG-91
MSc       3 Aditi     20   90      73       56 20-SEP-92
MCA       4 Sumit     20 57.5      78       81 07-DEC-91
MCA       5 Anirban   22   80      68       63 15-SEP-94
BCA       6 Kumkum    21   72    54.5       60 08-FEB-95
BCA       7 Suman     21 91.5      32       61 10-MAR-94
MSc       8 Rohit     22   85      76       92 19-APR-92
8 rows selected.
```

Q17: Update the Math marks of the student with Roll no 7 from 91.5 to 95 in the Student table.

UPDATE Student SET Math = 95 WHERE Roll = 7;

Output:

```
1 row updated.


SELECT * FROM Student WHERE Roll = 7;


ROLL NAME     AGE COURSE MATH PHYSICS COMPUTER BIRTHDAY
---- -------- --- ------ ---- ------- -------- --------
   7 Suman     21 BCA      95      32       61 10-MAR-94
1 row selected.
```

Q18: Update the Name of the student with Roll no 4 from Sumit to Sumitava in the Student table.

UPDATE Student SET Name = 'Sumitava' WHERE Roll = 4;

Output:

```
1 row updated.

SELECT * FROM Student WHERE Roll = 4;

ROLL NAME     AGE COURSE MATH PHYSICS COMPUTER BIRTHDAY
---- -------- --- ------ ---- ------- -------- --------
   4 Sumitava  20 MCA    57.5      78       81 07-DEC-91
1 row selected.
```

Q19: Delete the details of the student with Roll no 2 from the Student table.

DELETE FROM Student WHERE Roll = 2;

Output:

```
1 row deleted.

SELECT * FROM Student;

ROLL NAME     AGE COURSE MATH PHYSICS COMPUTER BIRTHDAY
---- -------- --- ------ ---- ------- -------- --------
   1 Rahul     19 BCA    79.5      67       89 15-JUN-93
   3 Aditi     20 MSc      90      73       56 20-SEP-92
   4 Sumitava  20 MCA    57.5      78       81 07-DEC-91
   5 Anirban   22 MCA      80      68       63 15-SEP-94
   6 Kumkum    21 BCA      72    54.5       60 08-FEB-95
   7 Suman     21 BCA      95      32       61 10-MAR-94
   8 Rohit     22 MSc      85      76       92 19-APR-92
7 rows selected.
```

Q20: Delete the details of all students from the Student table.

DELETE FROM Student; -- OR: TRUNCATE TABLE Student;

Output:

```
Note: run after Q19, 7 rows remain in the table.

DELETE FROM Student;
7 rows deleted.

-- OR --
```

```
TRUNCATE TABLE Student;
Table truncated.

SELECT * FROM Student;
no rows selected
```